Foundations of AI & ML **Fall 2026**
Professor Amy Greenwald[1] **Supervised Learning Unit**

Bias & Regularization: Ridge Regression & LASSO

1 Polynomial Regression

When $d = 1$, a **linear model** is a model in which each y_i is approximated by $mx_i + b$. Here, as in high school, m is the slope, while b is the y intercept, which shifts the line, and serves as the **bias** term in regression.

It is straightforward to incorporate a y intercept, and its accompanying bias, into our matrix notation. The trick is to append an extra dimension to the parameter vector $\boldsymbol{w} \in \mathbb{R}^d$, and to likewise append an extra feature/column to X, whose value is always 1, so that w_{d+1} represents the intercept:

$$X\boldsymbol{w} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} & 1 \\ x_{21} & x_{22} & \dots & x_{2d} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nd} & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_{d+1} \end{bmatrix} = \begin{bmatrix} w_1x_{11} + w_2x_{12} + \ldots + w_dx_{1d} + w_{d+1} \\ w_1x_{21} + w_2x_{22} + \ldots + w_dx_{2d} + w_{d+1} \\ \vdots \\ w_1x_{n1} + w_2x_{n2} + \ldots + w_dx_{nd} + w_{d+1} \end{bmatrix}$$

In a simple regression of y on $X = \begin{bmatrix} \boldsymbol{x} \end{bmatrix}$, i.e., only one feature, the aforementioned trick is akin to appending $\boldsymbol{x}^0 = \mathbf{1}$ to X, resulting in $\begin{bmatrix} \boldsymbol{x} & 1 \end{bmatrix}$, or $\begin{bmatrix} 1 & \boldsymbol{x} \end{bmatrix}$. More generally, it is equally possible to append $\boldsymbol{x}$ raised to any other power p to X as well: e.g., $\begin{bmatrix} 1 & \boldsymbol{x} & \boldsymbol{x}^2 & \boldsymbol{x}^3 & \dots & \boldsymbol{x}^p \end{bmatrix}$. Therefore, **polynomial regression** generalizes linear regression! In theory, at least. In practice, the set of possible combinations of powers of features is massive. Assuming just two features, say $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ and $p = 3$ yields the following (long) list of possibilities:[2]

$$1 \;\; \boldsymbol{x}_1 \;\; \boldsymbol{x}_1^2 \;\; \boldsymbol{x}_1^3 \;\; \boldsymbol{x}_2 \;\; \boldsymbol{x}_2^2 \;\; \boldsymbol{x}_2^3 \;\; \boldsymbol{x}_1\boldsymbol{x}_2 \;\; \boldsymbol{x}_1^2\boldsymbol{x}_2 \;\; \boldsymbol{x}_1\boldsymbol{x}_2^2 \;\; \boldsymbol{x}_1^2\boldsymbol{x}_2^2 \;\; \boldsymbol{x}_1^3\boldsymbol{x}_2 \;\; \boldsymbol{x}_1\boldsymbol{x}_2^3 \;\; \boldsymbol{x}_1^3\boldsymbol{x}_2^2 \;\; \boldsymbol{x}_1^2\boldsymbol{x}_2^3 \;\; \boldsymbol{x}_1^3\boldsymbol{x}_2^3$$

Feature selection is an important and difficult problem in machine learning. The deep learning revolution can in large part be attributed to its success at automatically identifying pertinent features.

2 Regularized Regression

Linear regression is a machine learning model with a strong bias, namely the decision to fit a *line* (as opposed to a curve) to data. Even with this inherent bias, linear regression models can have high variance. Another way to limit variance is to limit the range of the model parameters $\boldsymbol{w} \in \mathbb{R}^d$.

The p-**norm** of a vector $\boldsymbol{w} \in \mathbb{R}^d$, denoted $\|\boldsymbol{w}\|_p$, for some $p \geq 1$, is a way to gauge $\boldsymbol{w}$'s size.[3]

[1] This notes were compiled in conjunction with Professor Eric Ewing.

[2] I probably missed some!

[3] A norm is a function from $\mathbb{R}^d \to \mathbb{R}$ that satisfies the following three properties:

1. $\|\boldsymbol{x}\| > 0$, for all $\boldsymbol{x} \neq 0$
2. $\|\alpha\boldsymbol{x}\| = |\alpha|\|\boldsymbol{x}\|$, for all $\alpha \in \mathbb{R}$ and $\boldsymbol{x} \in \mathbb{R}^d$
3. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$, for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$

The final property is called the **triangle inequality**, because the sum of the lengths of two sides of a triangle is at least that of the third.

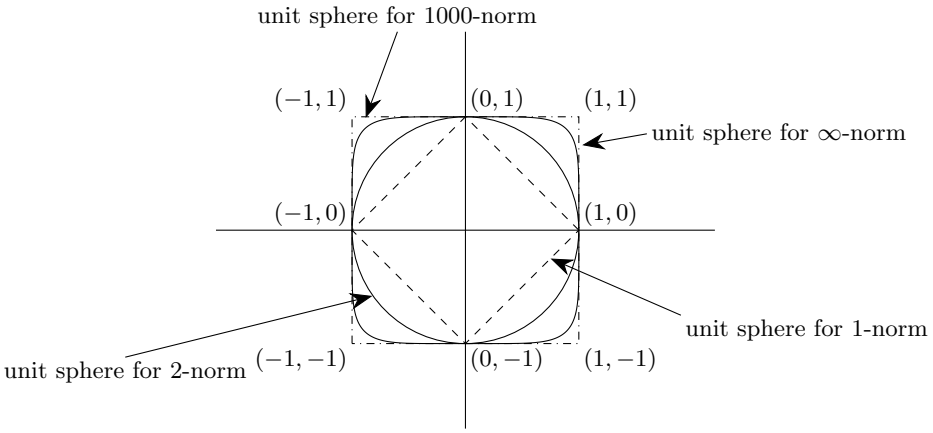


Figure 1: The feasible sets when the p-norm of a vector in $\mathbb{R}^d$ is constrained to be less than or equal to 1, for $p \in \{1, 2, 1000, \infty\}$. Original image source.

The most popular norm is the 2-norm, also called the **Euclidean norm**:

$$\|\boldsymbol{w}\|_2 = \sqrt{\sum_{i=1}^{d} w_i^2}$$

Other common choices include the 1-norm:

$$\|\boldsymbol{w}\|_1 = \sum_{i=1}^{d} |w_i|$$

and the ∞-norm:

$$\|\boldsymbol{w}\|_\infty = \max_{i \in \{1,\ldots,d\}} |w_i|$$

Figure 1 depicts the feasible sets when the p-norm of a vector in $\mathbb{R}^d$ is constrained to be less than or equal to 1, for $p \in \{1, 2, 1000, \infty\}$.

Recall the ordinary least squares (OLS) objective:

$$\text{loss}(\boldsymbol{w}) = (\boldsymbol{y} - X\boldsymbol{w})^T(\boldsymbol{y} - X\boldsymbol{w}) = \sum_{i=1}^{n} (\boldsymbol{y} - X\boldsymbol{w})_i^2$$

In other words,

$$\text{loss}(\boldsymbol{w}) = \|\boldsymbol{y} - X\boldsymbol{w}\|_2^2$$

Observe that OLS is an unconstrained optimization problem, as $\boldsymbol{w}$ can take on any value in $\mathbb{R}^d$. **Regularized regression** is a constrained optimization problem, with the same objective, but a limited domain for $\boldsymbol{w}$.

Ridge regression uses the 2-norm as a regularizer: for some $\beta > 0$,

$$\min_{\boldsymbol{w} \in \mathbb{R}^d} \|\boldsymbol{y} - X\boldsymbol{w}\|_2^2$$
$$\text{subject to } \|\boldsymbol{w}\|_2 \leq \beta$$

LASSO uses the 1-norm: for some $\beta > 0$,

$$\min_{\boldsymbol{w} \in \mathbb{R}^d} \|\boldsymbol{y} - X\boldsymbol{w}\|_2^2$$
$$\text{subject to } \|\boldsymbol{w}\|_1 \leq \beta$$

The choice of β is a choice of how much bias to include in the model.

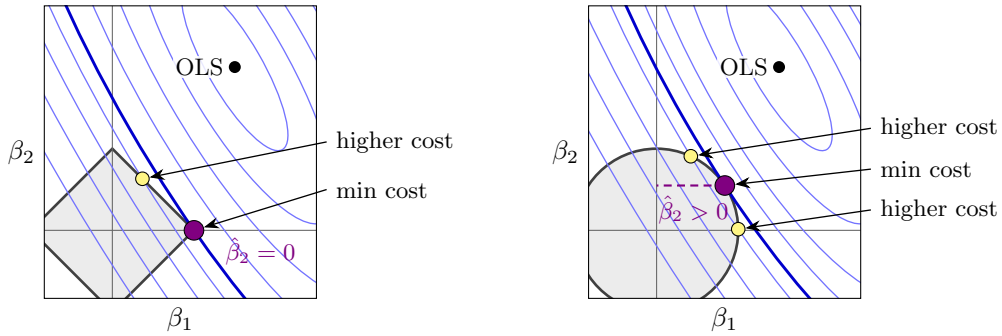
Like OLS, ridge regression has a closed-form solution. This solution can be found by first reformulating the ridge regression constrained optimization problem in terms of its Lagrangian,[4] and then following the same steps as last time to solve OLS. The closed-form solution of ridge regression generalizes that of OLS:

$$\boldsymbol{w} = (X^T X + \lambda I)^{-1} X^T \boldsymbol{y}$$

On the other hand, LASSO does not have a closed-form solution. The difficulty arises from the fact that the absolute value function is not differentiable at zero. As a result, LASSO can be solved using a generalization of gradient descent called *subgradient* descent, as the subgradient exists everywhere. The subgradient, however, is not unique, so this algorithm is usually slower to converge than gradient descent.[5]

While both ridge regression and LASSO impose bias, and thereby limit variance, their behavior is notably different. Take, for example, the vectors $(1, 0)$ and $(1/\sqrt{2}, 1/\sqrt{2})$. While $\|(1,0)\|_2 = \|(1/\sqrt{2}, 1/\sqrt{2})\|_2 = 1$, only $(1, 0)$ has 1-norm 1; $\|(1/\sqrt{2}, 1/\sqrt{2})\|_1 = 2/\sqrt{2} = \sqrt{2}$. As a result, **LASSO has a tendency to select features**, by completely zeroing out less important features, not just assigning them small coefficients.

Figure 2 is a depiction of ridge regression and LASSO. The OLS minimum is the dark black dot. As this point lies outside the feasible set, the optimal point for each problem lies on the boundary of the feasible set. But only for LASSO does it fall on a corner; the coefficient in the y direction is completely zeroed out.



ℓ_1 **constraint (LASSO)** ℓ_2 **constraint (Ridge)**

Figure 2: A depiction of the contour map for an OLS problem, as well as the feasible regions the LASSO and Ridge regression variants. There are an infinite number of contour lines; the figure depicts a few of them, including the ones that meet the respective feasible regions. LASSO prefers a zero coefficient for $\hat{\beta}_2$, while Ridge favors a non-zero setting. Original image source.

[4] The λ in this expression is called a Lagrange multiplier, and can be interpreted as a penalty for violating the constraint.

[5] In practice, LASSO problems are in fact solved using an algorithm called **coordinate descent**, which proceeds, for example, in a round-robin fashion, optimizing one coordinate at a time.